# Exp. No: 4 Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

## OBJECTIVES

1. To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.

2. To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).

3. To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

## SOFTWARE / HARDWARE REQUIRED

• OptiSystem Software

## EXPERIMENT PROCEDURE

1.Start a new OptiSystem project.

2. Place a User Defined Bit Sequence Generator and set the global bit rate to 40 Gb/s.

3. Place an Optical Gaussian Pulse Generator (193.1 THz, 1 mW, 12.5 ps initial pulse, 0.5 times bit slot) and connect it to the bit sequence generator.

4. Connect an Optical Time Domain Visualizer and an Optical Spectrum Analyzer at the generator output to observe the initial pulse.

5. Connect the generator output to the SMF and set the fiber length to 10 km.

6. Connect a visualizer and a spectrum analyzer at the SMF output to observe the broadened pulse.

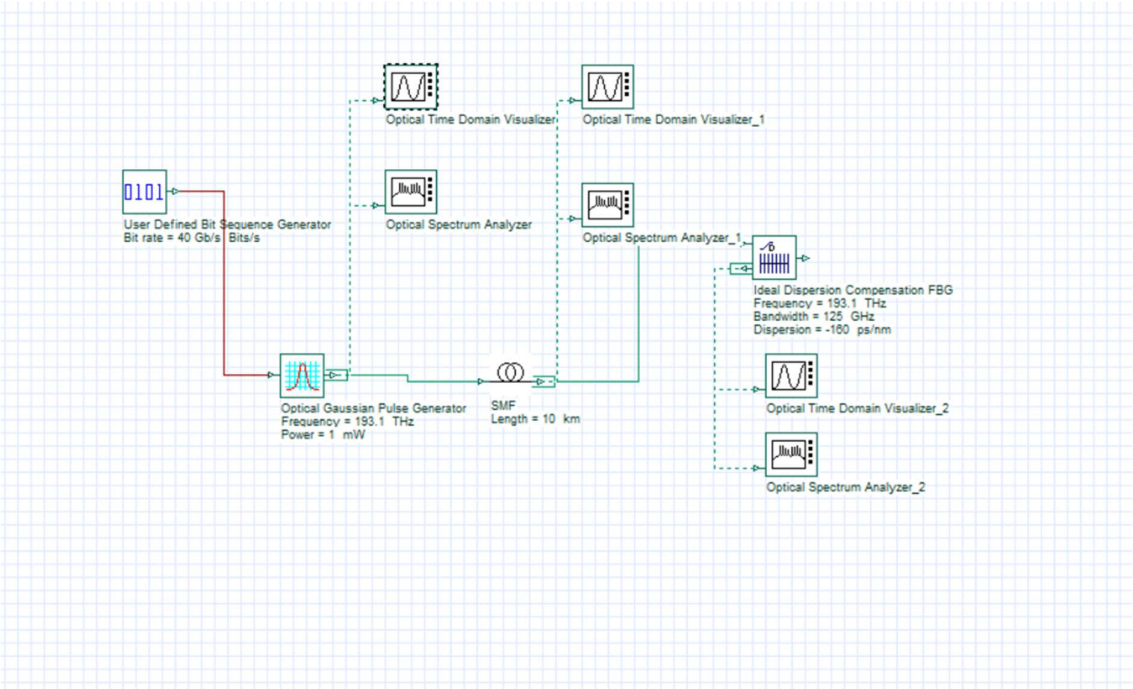
7. Connect the SMF output to the Ideal Dispersion Compensation FBG; in the Main tab set frequency = 193.1 THz, bandwidth = 125 GHz and dispersion = – 160 ps/nm.

8. Connect a visualizer and a spectrum analyzer at the compensator output.

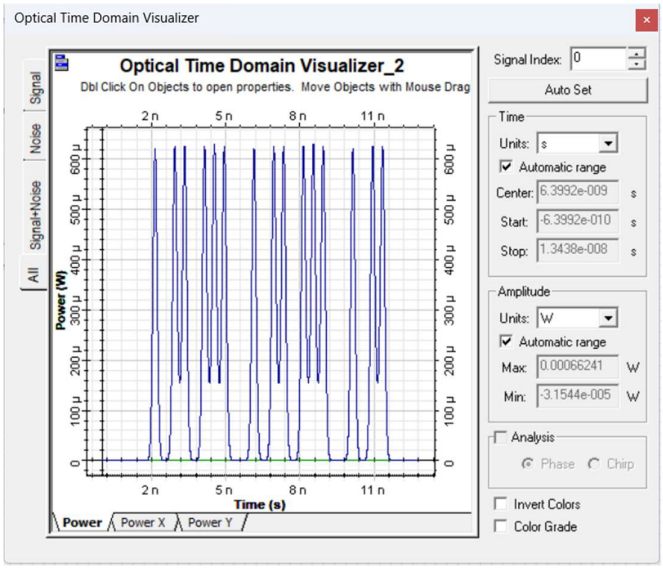
9. Run the simulation.

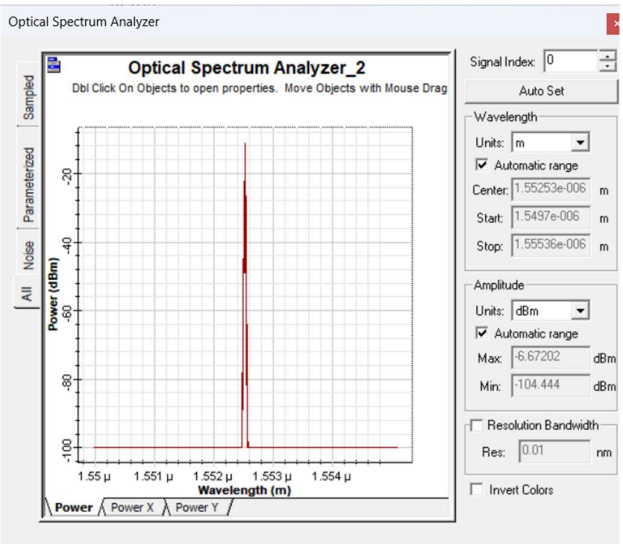
10. Compare the pulse width before the fiber, after 10 km of SMF and after dispersion compensation.

# DISPERSION COMPENSATION BLOCK DIAGRAM




# OUTPUT






**LABORATORY PERFORMANCE ASSESSMENT RUBRIC**

| Objective | Assessment Criterion | Excellent – 4 | Good – 3 | Satisfactory – 2 | Needs Improvement – 1 | Weightage |
|---|---|---|---|---|---|---|
| Design the OptiSystem layout | Project layout | Builds the complete layout with correct components and connections independently | Minor connection errors corrected | Needs guidance for components or connections | Unable to build the layout | 20 |
| Configure pulse, SMF and compensator parameters | Parameter configuration | Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, –160 ps/nm, 125 GHz) correctly | Minor parameter errors | Needs guidance for several parameters | Unable to configure parameters | 20 |
| Observe pulse broadening in SMF | Pulse observation | Captures and interprets initial and broadened pulses correctly | Minor interpretation errors | Limited observation / needs help | Unable to obtain or interpret pulses | 20 |
| Verify compensation using ideal dispersion component | Compensation verification | Verifies recovery of the pulse and explains residual effects of settings | Mostly correct verification | Partial verification with assistance | Unable to verify compensation | 20 |
| Document findings | Report and conclusion | Complete report with justified conclusion | Minor omissions | Incomplete record / weak conclusion | Report / conclusion inadequate | 20 |

## CONCLUSION

Dispersion compensation was successfully demonstrated using the Ideal Dispersion Compensation component in OptiSystem. A 12.5 ps Gaussian pulse broadened approximately four times after 10 km of SMF (accumulated dispersion 160 ps/nm), and sejing the compensator dispersion to –160 ps/nm r